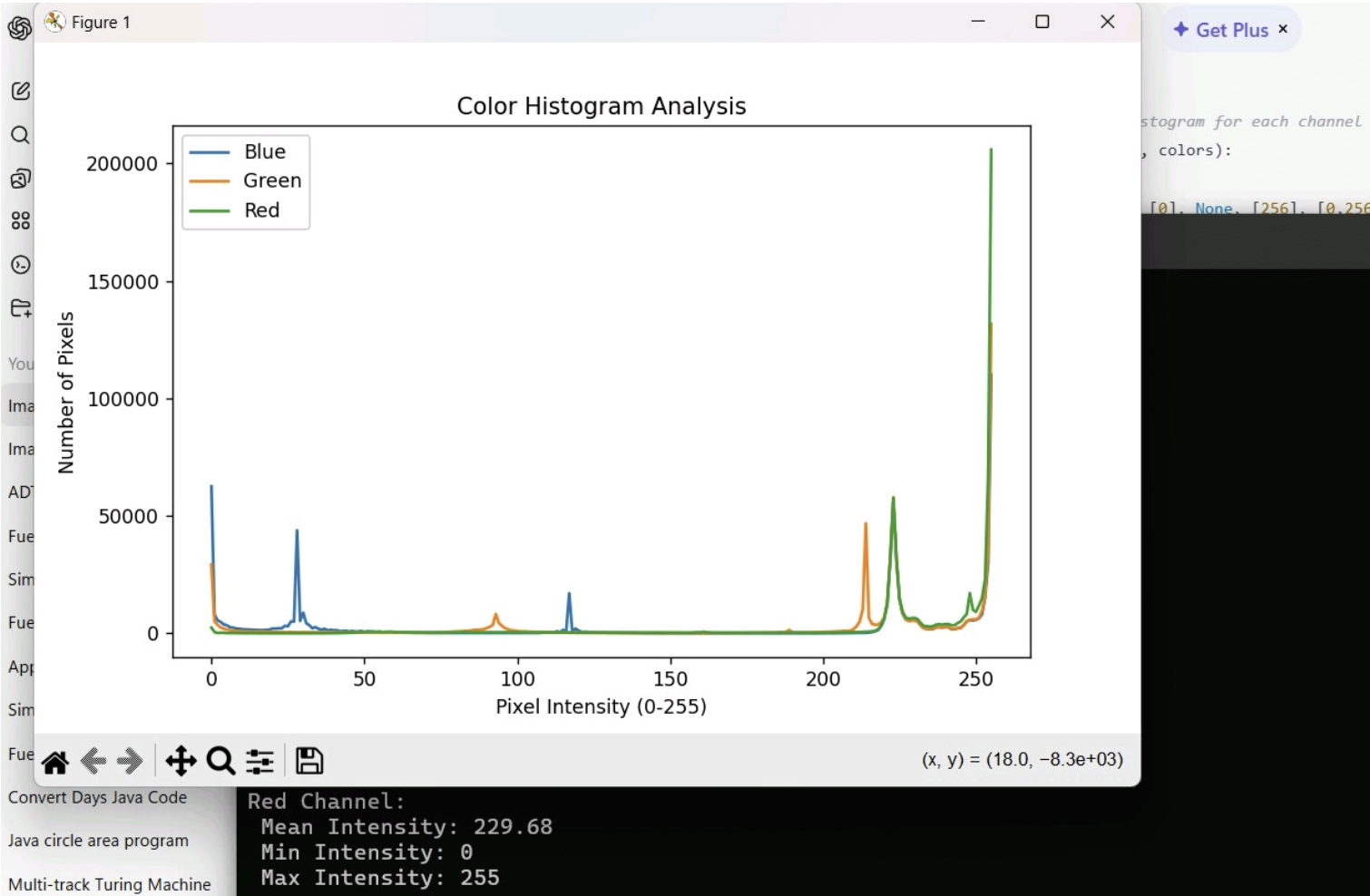






File Edit View

```
import cv2
import numpy as np
import matplotlib.pyplot as plt

def analyze_color_histogram(image_path):

    # Step 1: Read image
    image = cv2.imread(image_path)

    if image is None:
        print("Error: Image not found!")
        return

    # Step 2: Split into B, G, R channels
    channels = cv2.split(image)
    colors = ('Blue', 'Green', 'Red')

    plt.figure(figsize=(8,5))
    print("Color Level Analysis:\n")

    # Step 3: Calculate and analyze histogram for each channel
    for channel, color in zip(channels, colors):

        hist = cv2.calcHist([channel], [0], None, [256], [0,256])

        # Plot histogram
        plt.plot(hist, label=color)

        # Basic statistical analysis
        mean_intensity = np.mean(channel)
        min_intensity = np.min(channel)
        max_intensity = np.max(channel)

        print(f"{color} Channel:")
        print(f" Mean Intensity: {mean_intensity:.2f}")
        print(f" Min Intensity: {min_intensity}")
        print(f" Max Intensity: {max_intensity}")
        print("-"*30)
```